UNIVERSITY OF CAPE TOWN
Investment Decision Theory and Application- ECO4102Z/ECO4122Z
Tutorial 1

1. Endowments are funds that are typically owned by non-profit institutions involved in educational, medical, cultural, and other charitable activities. Classified as institutional investors, endowments are almost always established with the intent of lasting into perpetuity.
   The University of Cape Town Endowment was established in South Africa to provide financial support to University of Cape Town. An endowment's spending rate defines the fraction of endowment assets distributed to the supported institution. The University of Cape Town Endowment has established a spending rate of 4 percent a year; the endowment follows the simple rule of spending, in a given year, an amount equal to 4%× (Market value of the endowment at the end of the prior year). This amount is committed to the budgetary support of the university for the coming year. At the end of the prior year, the market value of the University of Cape Town Endowment's assets stood at R75,000,000 000. In addition, the University of Cape Town Endowment has committed to contribute R1,000,000,000 in the coming year to the construction of a new student residence. Planners at the endowment expect the endowment to receive contributions or gifts (from alumni and other sources) of R400,000,000 over the coming year. What is the anticipated liquidity requirement of the University of Cape Town Endowment for the coming year? [4 marks]

2. The Executive Director of the UCT University Endowment estimates that the capital markets will provide a 9 percent expected return for an endowment portfolio taking above-average risk, and a 7 percent expected return for an endowment portfolio taking average risk. The UCT Endowment provides tuition scholarships for UCT students. The spending rate has been 4 percent, and the expected tuition inflation rate is 3 percent. Recently university officials have pressured the endowment to increase the spending rate to 6 percent. The endowment has an average to below-average ability to accept risk and only an average willingness to take risk, but a university official claims that the risk tolerance should be raised because higher returns are needed. Discuss an appropriate return objective and risk tolerance for the UCT Endowment. [4 marks]

3. Characterize each of the investment objectives given below as one of the following: an absolute risk objective, a relative risk objective, an absolute return objective, or a relative return objective.
   1. Achieve a rate of return of 8 percent a year. ARE

1. Liq req

1B − 400m − ~~4% × 75B~~ 600m
= −2,4B ✗

A600m

∴ liq req = R2 400 000 000 ✗

2.

?	?	7%	9%	(E(R))
BA	A→BA	Ave	AA	

3% inflation
4% → 6% spending rate

ret. objective should be consistent with tolerance (Ave) ∴ should be 7%. 6% spending rate too high

Ability : BA → Ave
Willingness : Ave
risk tol. should be raised

risk tol limited by ability + willingness

Higher returns needed
∴ risk tolerance should adjust from Ave to Ave → AA to allow for the higher returns of 8% while not deviating too much from current profile

4 a) Life expectancy = 30 more years

1,35m + .18 × 2 + .27

infl = 6%

= 1,98m
÷ 50m × 100
= 3,96%
≈ 4,2% after inflation

b) High ability (mod - high) — No debt, grown children, 50m asset base (even left work to persue)
High willingness
High risk tolerance

RRC × ARI

2. Limit the standard deviation of portfolio returns to 20 percent a year or less.

ARC ✓

3. Achieve returns in the top quartile of the portfolio's peer universe (the set of portfolios with similar investment objectives and characteristics).

4. Maintain a 10 percent or smaller probability that the portfolio's return falls below the threshold level of 5 percent per annum over a one-year time horizon.

RRI ✓

[4 marks]

11:00

4. Lisa Henwood, a registered nurse and businesswoman, recently retired at age 50 to pursue a life as a blues singer. She had been running a successful cosmetics and aesthetics business. She is married to Ted, a retired scientist (age 55). They have saved R50 million in their portfolio and now they want to travel the world. Their three children are all grown and out of college and have begun their own families. Lisa now has two grandchildren. . Lisa and Ted feel that they have achieved a comfortable portfolio level to support their family's needs for the foreseeable future. 50M

1,35M

To meet their basic living expenses, Ted and Lisa feel they need R1,350,000 per year in today's terms (before taxes) to live comfortably. As a trained professional, Lisa likes to be actively involved in intensively researching investment opportunities. Lisa and Ted want to be able to provide R180,000 per year (pretax) indexed for inflation to each of their grandchildren over the next 10 years for their college education. They also want to set aside R270,000 each year (pretax) indexed for inflation for traveling for her musical performances around the South Africa. They have no debt. Most of their portfolio is currently in large-cap RSA stocks and Treasury notes.

willing

safer - more liquid

They have approached Rele Tsoai, CFA, for guidance on how to best achieve their financial goals. Inflation is expected to increase at an annual rate of 6 percent into the foreseeable future.

(a) What is Lisa's return objective? [2 marks]

(b) Determine Lisa's willingness and ability to assume risk and based on this, deduce their risk tolerance level. [3 marks]

(c) Based on the information in the case, which one of the following portfolios should the Lisa choose? [5 marks]

Question 4 continues on the next page. . .

Asset Class	Expected Return	Allocation		
		Portfolio A	Portfolio B	Portfolio C
RSA large stocks	9%	20%	5%	10%
RSA small stocks	10	20	15	10
Foreign stocks	12	15	15	10
Corp. bonds	5	15	0	35
Gov. bonds	3	10	0	25
Venture capital	11	5	30	0
REITs	15	10	30	0
Cash	1	5	5	10
Pretax return		11.8%	14.6%	8.7%
After-tax return	gains + income	8.6%	10.4%	6.6%
After-tax yield	div + coupons	4.9%	4.9%	5.8%